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Clinical review of robotic distal pancreatectomy

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BACKGROUND

- Distal pancreatectomy has shifted from open (ODP) to minimally invasive surgery: laparoscopic (LDP, since 1994) and robotic (RDP).
- LDP is established but limited by **dexterity and visualisation**.
- RDP addresses these limits, yet whether it improves outcomes over LDP remains unsettled.

AIM A clinical review comparing RDP with ODP and LDP: technique, clinical outcomes, cost-effectiveness and challenges.

OPERATIVE TECHNIQUE

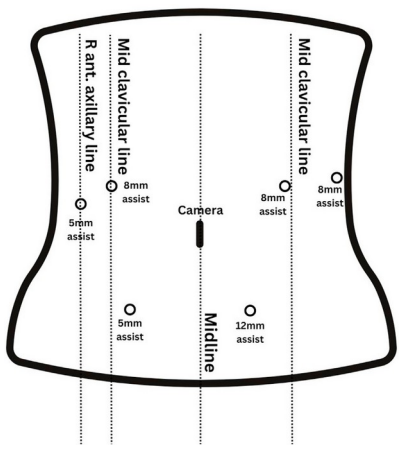


Fig. 1 Port placement: 8 mm robotic arms, 12 mm camera, 5 and 12 mm assistant ports

1 Position Supine or split legs; steep reverse Trendelenburg, then dock

2 Expose Enter the lesser sac to expose the pancreas

3 Control Splenic artery and splenic vein

4 Divide Pancreas with a macroscopic margin

5 Resect En bloc retroperitoneal dissection ± spleen

Spleen-preserving options: splenic vessel preservation or Warshaw technique

OTHER ROBOTIC PLATFORMS

Platform	Distal pancreatectomy data	Notes
da Vinci Intuitive	Almost all published RDP	Reference platform
Hugo Medtronic	No cases reported	First generation
Versius CMR	9-case series	Mean OT 185 min, LOS 7 d; slower set-up, no integrated energy device
hinotori™ Medicaroid	Single case report	Docking-free, flexible arms; longer set-up, no vessel sealer

CHALLENGES

Operative time
≈30 min longer than LDP in meta-analysis, largely robot docking; shortens along the learning curve.

Team & training
Needs a console surgeon, skilled bedside assistant and robot-trained scrub nurse; few training centres.

Cost & access
High upfront cost; robots concentrated in advanced economies, creating inequity of access.

WHY ROBOTIC?

3D magnified visualisation

Tremor filtration

Motion scaling

Wristed articulating instruments

CLINICAL OUTCOMES OF RDP

Outcome	RDP vs ODP	RDP vs LDP
Blood loss	▼ Lower	▼ Lower (planned SP)
Complications	▼ Fewer (fistula, respiratory, wound)	= Similar overall, major & fistula
Hospital stay	▼ Shorter	▼ ≈1 day shorter
Conversion to open	n/a	▼ Lower
Spleen preservation	▲ Higher	▲ Higher; fewer conversions to splenectomy
Oncological	= Non-inferior R0 and LN yield (MIDP)	▲ 0% vs 2% R1; more lymph nodes
Operative time	= Conflicting	▲ ≈30 min longer (docking)

Green favours RDP **Grey** no clear difference **Amber** favours comparator. SP, spleen preservation; MIDP, minimally invasive DP

1 day shorter stay vs LDP

0% positive margins (LDP 2%)

+30 min operative time vs LDP

COST-EFFECTIVENESS

Operative cost **RDP > LDP, ODP**

Overall cost **RDP ≤ LDP < ODP**

QALYs **RDP > LDP**

- Higher instrument and platform costs are **offset by shorter stay, fewer conversions and lower morbidity**.
- Costs fall with case volume (learning curve, robot-specific instruments).
- Higher QALYs than LDP at similar cost, within the European acceptability threshold; supported by the multicentre ROBOCOSTES trial.

CONCLUSION

RDP is a safe, cost-effective advance in distal pancreatectomy, offering greater precision, better clinical outcomes and higher spleen preservation than open and laparoscopic approaches.

Wider adoption depends on expanding training, dedicated robotic teams and equitable access.

Hwang N, Reddish S, Koea J, Srinivasa S. Clinical review of robotic distal pancreatectomy. *Laparoscopic, Endoscopic and Robotic Surgery* 2025;8:80–83. doi:10.1016/j.lers.2025.02.001